

Spotify User Churn Analysis Using SQL

Project Objective

Analyze Spotify user behavior and identify factors influencing customer churn using SQL.

Dataset Description

The dataset contains user information such as User ID, Country, Subscription Type, Device Type, Listening Time, Songs Played Per Day, Ads Listened Per Week, Skip Rate, and Churn Status.

SQL Concepts Used

Aggregate Functions, GROUP BY, HAVING, CASE WHEN, Subqueries, CTEs, Views, ROW_NUMBER(), RANK(), DENSE_RANK().

Key Analyses Performed

- Average Listening Time by Subscription Type
- Average Skip Rate by Subscription Type
- Device-wise Listening Time
- Total Churned Users
- Churn Rate by Subscription Type
- Churn Rate by Device Type
- Active vs Churned User Analysis
- Users with Above Average Listening Time
- Users with Above Average Skip Rate
- Top Listeners Analysis
- Country-wise Listening Time Analysis
- Country-wise Churn Analysis

View Created

churned_summary view created to summarize churned users by subscription type.

Key Insights

1. Free users showed the highest churn rate.
2. Student users showed the lowest churn rate.
3. Mobile users had better retention than Web users.
4. Higher skip rates were associated with churned users.
5. Active users generally had higher listening time.
6. User engagement metrics can help identify churn risk.

Conclusion

This project demonstrates the use of SQL for customer churn analysis, user behavior analysis, and business insight generation using real-world analytical techniques.